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Supplementary Information: Martian Crustal Magnetic Field Heterogeneity and Implications for Biological Systems at Candidate Landing Sites

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Extended Methods

Spherical Harmonic Model and Vector Field Processing

Martian crustal magnetic field components (B_r, B_θ, B_ϕ) were computed using the spherical harmonic potential model developed by Langlais et al.[1], derived from joint inversions of nightside orbital data from Mars Global Surveyor (MGS) at 400 km circular mapping altitude and MAVEN elliptical periapsis tracks down to $\sim$ 130 km.

Ground-Truth Comparison with InSight Magnetometer

Direct in situ ground-truth validation was integrated using calibrated surface magnetometer measurements from the InSight Fluxgate (IFG) Magnetometer in Elysium Planitia (4.50°N, 135.62°E). The InSight surface measurement of $\sim$ 2013 nT[2] confirms that shallow, localized crustal sources in the subsurface can produce ground-level fields an order of magnitude higher than orbital baseline projections.

Supplementary Figures

Data and Code Availability

The full machine-readable landing site database, processed global grids, figure generation scripts, and interactive 3D WebGL globe source code are available in the public GitHub repository (`https://github.com/dr-richard-barker/mars-magnetic-biology`) and archived under DOI: 10.5281/zenodo.XXXXXXX.

References

1. B. Langlais, E. Thébault, B. Sanchez-Cano, J. E. P. Connerney, A. Mittelholz, and C. L. Johnson. A new model of the crustal magnetic field of mars using MAVEN and Mars Global Surveyor data. *Journal of Geophysical Research: Planets*, 124(6):1542–1569, 2019. doi: 10.1029/2019JE005979.

2. C. L. Johnson, A. Mittelholz, B. Langlais, C. T. Russell, V. Ansan, D. Banfield, P. J. Chi, M. O. Fillingim, F. Forget, A. Gao, S. Goossens, M. Grott, J. Halekas, E. Hauber, D. Jault, B. Knapmeyer-Endrun, M. Lemmon, N. Mueller, M. Panning, S. E. Smrekar, A. Spiga, R. Weber, M. A. Wieczorek, and W. B. Banerdt. Crustal and time-varying magnetic fields at the InSight landing site on Mars. *Nature Geoscience*, 13:199–204, 2020. doi: 10.1038/s41561-020-0536-x.

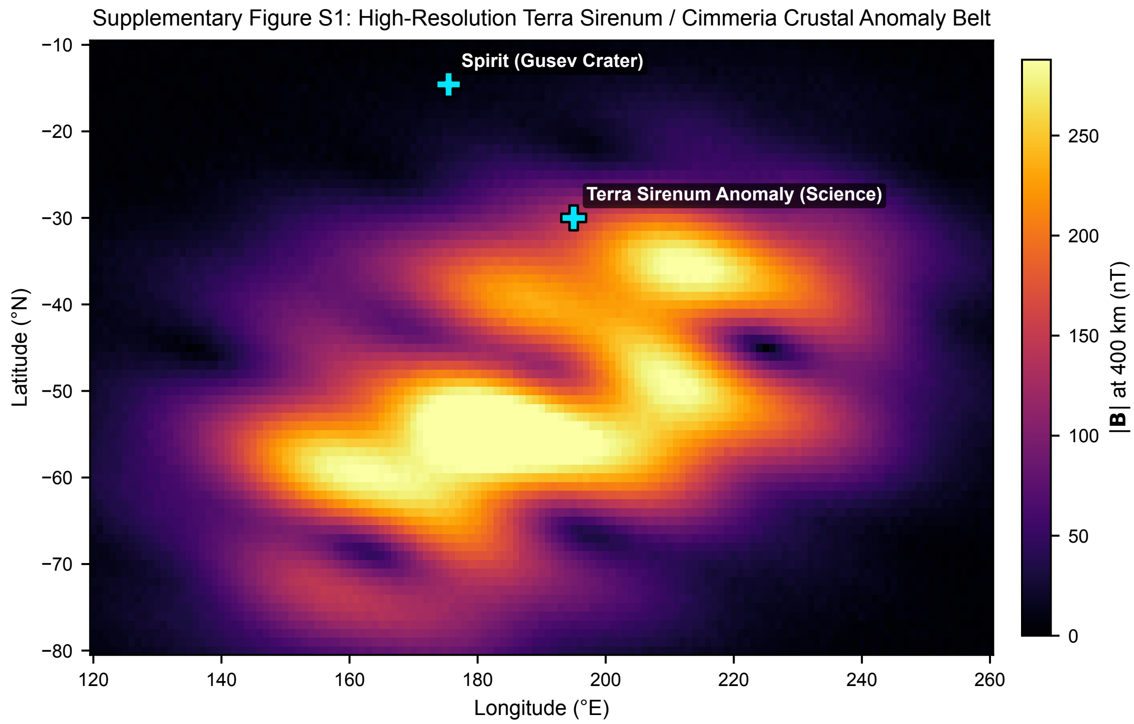

Figure 1 | **Supplementary Figure S1 | High-resolution regional magnetic field map of the Terra Sirenum / Terra Cimmeria crustal anomaly belt.** Map spans 120°E to 260°E longitude and 10°S to 80°S latitude at 400 km altitude, highlighting coherent crustal magnetic lineations and candidate scientific outpost coordinates.

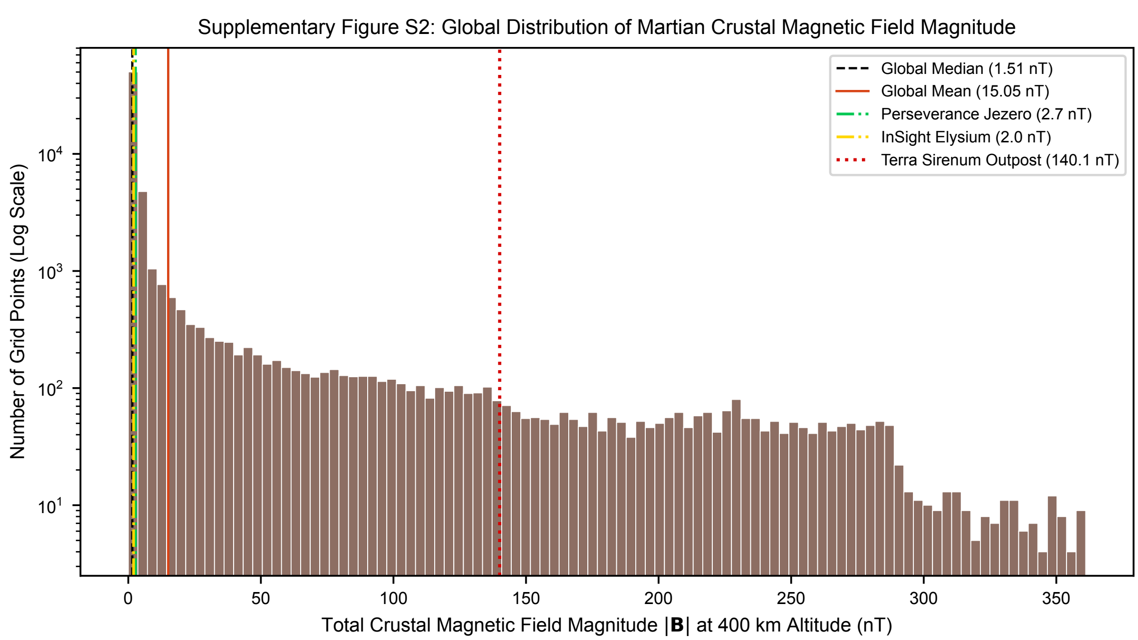

Figure 2 | **Supplementary Figure S2 | Global frequency distribution of Martian crustal magnetic field magnitude.** Histogram displaying |**B**| at 400 km altitude across 65,160 global grid points, with global median (1.51 nT), global mean (15.05 nT), Perseverance Jezero site (1.1 nT), InSight Elysium site (1.2 nT), and Terra Sirenum outpost (214.8 nT) highlighted.